
Anisotropic Manifold Packing for Self-Supervised Learning

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Abstract

Self-supervised vision learning has been largely based on contrastive comparisons between individual embedding vectors. In the brain’s cortex, responses to distinct classes are represented as geometric structures known as neural manifolds. Like in a packing problem, accurate classification can be achieved by separating these manifolds. We utilize **Contrastive Learning As Manifold Packing (CLAMP)**, which models each image’s augmented embeddings as an ellipsoidal sub-manifold and separates them. The sizes and positions of the sub-manifolds are optimized by following the gradient of a packing loss. However, CLAMP collapses each sub-manifold’s full covariance to a scalar radius, reducing oriented ellipsoids to *isotropic*, identical spheres for overlap detection and discarding directional information critical for identifying true geometric overlap. We propose Shape-aware manifold packing, which restores the discarded geometry through an anisotropic overlap criterion based on the Mahalanobis distance over combined sub-manifold covariances, and an asymmetric orientation-aware repulsion that attenuates force along each sub-manifold’s principal axes. Utilizing the standard linear evaluation protocol, our model achieves comparable performance with standard CLAMP, highlighting the potential of self-supervised learning.

1 Introduction

Learning visual representations that generalize across tasks without labeled data is a central goal of modern computer vision. Contrastive self-supervised learning addresses this problem by training encoders to pull together embeddings of augmented views of the same image while pushing apart embeddings of different images [1, 3, 2]. The geometry this process induces in feature space resembles structures in the brain called *neural manifolds* [4, 5]. Classification in these biological systems depends on how well-separated the manifolds are. When manifolds corresponding to different categories overlap in neural state space, the resulting ambiguity leads to misclassification. CLAMP [6] bridges physics, neural science, and machine learning by explicitly modeling each image’s augmented embeddings as sub-manifolds, where a *sub-manifold* corresponds to the embedding of a single data sample, and a *manifold* to an entire class. Despite CLAMP computing the full covariance matrix Λ_i for each sub-manifold, the framework collapses this information into a single scalar radius. Furthermore, all pairwise repulsive interactions are treated identically, so semantically related categories (e.g., two dog breeds) are repelled as aggressively as unrelated ones (e.g., a dog and an airplane), with no mechanism for hierarchical organization. We address the following:

- We introduce an *anisotropic* overlap detection criterion, utilizing the covariance matrix’s distinct properties, like elongation and orientation, based on the Mahalanobis distance computed over the combined covariance of sub-manifold pairs, preserving the full ellipsoidal geometry during repulsion.

- We propose an asymmetric orientation-aware repulsion that reduces the repulsive force along each sub-manifold’s principal axes, controlled by a factor of $\gamma \in [0, 1]$.
- We show that these extensions produce an embedding space in which sub-manifolds are more compactly separated along geometrically meaningful directions, while preserving the CLAMP’s framework, and account for how including more complex features changes the features within the cortex.
- We evaluate both extensions on CIFAR-100 [10] under a 100-epoch compute budget with CLAMP’s hyperparameters held fixed. The isotropic baseline outperforms our variants in this regime; we report the outcome unadjusted and analyze why the shape signal our method depends on is not yet informative at this horizon.

2 Related work

2.1 Contrastive Self-Supervised Learning

Modern SSL methods learn representations by contrasting augmented views of images without access to labels. SimCLR [1] demonstrated that a simple framework pairing large batch sizes with a learnable nonlinear projection head and strong augmentations could produce high-quality visual features, using the normalized temperature-scaled cross-entropy (NT-Xent) loss to push apart negative pairs within each batch. MoCo [3] addressed the batch-size dependency by maintaining a momentum-updated queue of negative embeddings, decoupling the number of negatives from the batch size and enabling contrastive learning with fewer computational resources. BYOL [2] showed that the explicit negative-pair mechanism is not strictly necessary, instead using an asymmetric architecture with a slow-moving target network to avoid representational collapse. Across these methods, individual embeddings are treated as point vectors and compared through pairwise losses. None explicitly models the geometric structure of the set of augmented embeddings arising from a single image, as CLAMP treats an image as its own class, operating on the entire augmentation sub-manifolds surrounding each input embedding.

2.2 CLAMP and Manifold Packing

Neuroscience research suggests that neural populations encode stimuli as low-dimensional geometric structures embedded in high-dimensional activity spaces. Cunningham and Yu [4] developed techniques for extracting these neural manifold surfaces from large-scale neural recordings and showed that the true number of degrees of freedom in neural activity is usually far smaller than the number of neurons that were recorded. Jazayeri and Ostojic [5] related this “true” dimensionality to the kinds of computations a brain region can support. Chung et al. [9] tied the geometry of their size, shape, and orientation directly to how easily a simple classifier can separate the categories they encode. Together, these results suggest that it is useful to think of a learned representation not as a bag of isolated points but as a collection of geometric shapes whose arrangement in space determines how well downstream tasks can be solved.

Zhang et al. [6] brought this view into SSL with CLAMP. Instead of comparing individual embedding vectors, CLAMP treats the augmented views of each image as a single ellipsoidal sub-manifold, described by a centroid and a covariance matrix. The training objective is a short-range repulsive potential that pushes overlapping sub-manifolds apart and leaves non-overlapping ones alone, borrowing from physics models of colloidal suspensions and jammed packings in which overlapping particles get displaced while isolated particles sit still. The result is a single, physically interpretable loss with a small number of meaningful hyperparameters (r_s controls how big each sub-manifold is allowed to be, m controls how many augmentations make up each sub-manifold). On standard benchmarks including ImageNet-100, CLAMP matches or outperforms point-based SSL methods while keeping the framework simpler: one projection head, one loss term, no queues or slow-moving target networks [6].

The piece of CLAMP that our work targets is its overlap criterion. CLAMP computes the full covariance matrix Λ_i for every sub-manifold—the object that encodes its shape, elongation, and orientation, but then compresses Λ_i to a single scalar radius by taking its trace. After that step, two sub-manifolds that happen to have the same total variance are treated identically, even if one is a thin ellipsoid pointing in a specific direction and the other is a round ball. In effect, CLAMP

models each sub-manifold as an oriented ellipsoid internally and then interacts with it as if it were a sphere. The geometric information needed for shape-aware repulsion already exists inside CLAMP’s pipeline; it is just not used at the point where repulsion is computed. Our method keeps CLAMP’s framework in place and refines only this overlap step so that the ellipsoidal geometry CLAMP has already computed is actually put to work.

2.3 Clustering-based self-supervised learning

A separate family of SSL methods introduces explicit group structure by clustering embeddings. DeepCluster [7] alternates between grouping features with k -means and using the resulting cluster IDs as pseudo-labels that supervise the next round of training, showing that unsupervised grouping alone can recover meaningful categories. SwAV [11] replaces the offline k -means step with a soft online assignment and additionally enforces that different augmentations of the same image are placed in consistent clusters. Both methods bake the notion of a “group” directly into the loss: embeddings are explicitly told which cluster they belong to.

Our work sits closer to CLAMP than to clustering-based methods. We never cluster the dataset or assign pseudo-labels. Instead, we let each sub-manifold’s own geometry moderate how strongly it repels its neighbors: when a sub-manifold is already elongated along a particular direction, we reduce the force it exerts along that direction by a factor $\gamma \in [0, 1]$ (defined in Section 3.1.2), so semantically similar images can sit closer together in the embedding space without being actively pulled apart. We do not claim this is a replacement for explicit clustering, nor do we introduce a separate clustering loss; whether the resulting representations recover the known superclass structure of datasets such as CIFAR-100 [10] is an empirical question we examine in Section 4.

3 Method

3.1 Description

Drawing on the physics of short-range repulsive particle systems and random organization dynamics, CLAMP’s loss behaves like the potential energy between soft particles. Training follows the gradient of this energy, and the embedding space progressively self-organizes into a jammed-packing-like configuration without ever comparing individual embedding vectors pairwise. CLAMP operates directly on sub-manifolds, treating each one as a single geometric object whose shape and position are both optimized. This is what distinguishes it from point-level contrastive methods such as SimCLR [1] or MoCo [3], which reason only about relationships between individual augmented embeddings.

CLAMP [6] processes a batch of b images by generating m augmented views per image, encoding them through a ResNet-18 [8] backbone f_θ , and projecting the resulting representations onto the unit hypersphere using a MLP projection head g_ϕ . After projection, the augmented embeddings of image x_i form the sub-manifold $\{z_i^1, \dots, z_i^m\} \subset \mathbb{R}^D$, which CLAMP approximates as a high-dimensional ellipsoid. A centroid and an unbiased sample covariance are created where $Z_i \in \mathbb{R}^{m \times D}$ is the matrix of centered embeddings, $(Z_i)_{k,:} = z_i^k - \bar{z}_i$, and the covariance Λ_i encodes the full shape, elongation, and orientation of the ellipsoidal sub-manifold through its eigenvalues and eigenvectors.

$$\bar{z}_i = \frac{1}{m} \sum_{k=1}^m z_i^k, \quad \Lambda_i = \frac{1}{m-1} Z_i^\top Z_i \quad (1)$$

The trace operator then sums all eigenvalues of Λ_i , collapsing the full eigenspectrum into a single number. For instance, $\Lambda_A = \text{diag}(9, 1)$ and $\Lambda_B = \text{diag}(5, 5)$ both yield $\text{Tr} = 10$, despite having fundamentally different overlap behavior.

A loss function is used that penalizes overlap between all pairs of sub-manifolds using the Euclidean distance $d_{ij} = \|\bar{z}_i - \bar{z}_j\|$ and the scalar radius. We define $\mathcal{L}_{\text{overlap}}$ as the pairwise overlap loss and include a size regularization penalty to prevent sub-manifolds from expanding without bound, within

the full CLAMP training objective:

$$\mathcal{L}_{\text{CLAMP}} = \log\left(\mathcal{L}_{\text{overlap}} + \lambda_0 \sum_i r_i + \varepsilon\right), \quad (2)$$

$$\mathcal{L}_{\text{overlap}} = \sum_{i \neq j} \max\left(0, 1 - \frac{d_{ij}}{r_i + r_j}\right)^p, \quad (3)$$

where $p \geq 1$ controls the steepness of the repulsive potential (we set $p = 2$, matching CLAMP), $\lambda_0 \geq 0$ is the size-penalty weight, and ε is a small constant added for numerical stability of the logarithm. The logarithm of $\mathcal{L}_{\text{overlap}}$ compresses large values during early training when many pairs overlap. After pretraining, the MLP projection head g_ϕ is discarded, and a linear classifier is trained on the frozen backbone representations produced by f_θ .

Equation 3 makes repulsion *directionless*: two ellipsoids pointing into each other receive the same force as two sitting side by side, because the interaction depends only on centroid separation and a scalar radius. We propose two extensions that address this limitation. Both are designed to reduce to the original CLAMP behavior as special cases, ensuring that our objective is a strict generalization of Equation 2.

3.1.1 Anisotropic overlap detection

Our first extension replaces the standard Euclidean distance in Equation 3 with a *Mahalanobis distance* that accounts for the shape and orientation of each sub-manifold, computed under the *combined* covariance of each sub-manifold pair. For sub-manifolds i and j , we define this combined covariance as

$$\Sigma_{ij} = \Lambda_i + \Lambda_j \quad (4)$$

and the anisotropic distance

$$d_{\text{aniso}}(i, j) = \sqrt{(\bar{z}_i - \bar{z}_j)^\top \Sigma_{ij}^+ (\bar{z}_i - \bar{z}_j)}, \quad (5)$$

where Σ_{ij}^+ denotes the Moore–Penrose inverse of Σ_{ij} , allowing us to compute distances specifically in the subspace where the data actually lies. This distance behaves differently depending on direction. Intuitively, $d_{\text{aniso}}(i, j)$ is small when the separation $\bar{z}_i - \bar{z}_j$ aligns with large combined variance and large when the separation aligns with small combined variance. It measures whether two ellipsoids actually intrude on one another, given their shapes and orientations, rather than merely whether their centers are close.

When all sub-manifolds are isotropic, $\Lambda_i = \sigma^2 I$, Equation 5 reduces to $\|\bar{z}_i - \bar{z}_j\|/(\sigma\sqrt{2})$, recovering the Euclidean distance of the original CLAMP overlap criterion up to a global rescaling that can be absorbed into r_s .

3.1.2 Asymmetric orientation-aware repulsion

Replacing the distance metric alone does not fully exploit the available geometric information. Equation 5 changes the criteria for overlap, but the gradient induced by the resulting loss still pushes centroids apart with equal magnitude along every direction in $\mathbb{R}^D$. These directions are precisely the ones its augmented views already populate, so enforcing repulsion distorts the natural spread the encoder has learned and inflates r_i .

We therefore modify the backward pass of the packing loss to reduce the centroid gradient along each sub-manifold’s directions, rather than modifying the forward loss itself. Let $V_i \in \mathbb{R}^{D \times k}$ denote the singular k vectors of matrix Z_i , with $k \leq \text{rank}(\Lambda_i) \approx m$. These are the eigenvectors of Λ_i associated with nonzero eigenvalues. Let $P_i = V_i V_i^\top$ denote the orthogonal projector onto the subspace spanned by the principal directions of Λ_i . For a single scalar $\gamma \in [0, 1]$, we replace centroid gradient $g_i = \partial \mathcal{L}_{\text{aniso}} / \partial \bar{z}_i$ with

$$\tilde{g}_i = g_i - (1 - \gamma) P_i g_i = \gamma P_i g_i + (I - P_i) g_i. \quad (6)$$

Equivalently, $\tilde{g}_i$ leaves the areas of low variance unchanged, while scaling principal directions by γ . The loss itself is unchanged during the forward pass; only the gradient with respect to the

centroid is modified before being backpropagated through the projection head g_ϕ and the backbone f_θ . This is implemented using a custom autograd operation that behaves like the identity during the forward pass, but applies the gradient transformation defined in Equation 6 during the backward pass (see `GradientAttenuator` in `model/loss_module.py`). Since each V_i is computed from cluster i 's own centered views, every sub-manifold responds according to its own geometry, making the repulsion asymmetric by design.

Setting $\gamma = 1$ makes Equation 6 the identity and recovers anisotropic overlap detection without orientation-aware attenuation; additionally restricting Λ_i to be isotropic then exactly recovers the original CLAMP loss (Eq. 2). Setting $\gamma = 0$ removes all spread-axis repulsion, allowing two sub-manifolds whose long axes are aligned to coexist without force along that shared direction. Intermediate values let the model maintain separation along the compact axes, whose overlap is most harmful to classification, without distorting the natural spread of each sub-manifold.

3.1.3 Properties of the shape-aware packing loss

Shape-aware overlap detection. The anisotropic distance in Equation 5 measures geometric intrusion rather than centroid proximity. Two elongated sub-manifolds whose long axes point in different directions can have close centroids without meaningfully intersecting; conversely, two sub-manifolds whose long axes are aligned can overlap heavily even when their centroids are farther apart than the sum of their scalar radii. CLAMP cannot distinguish these two cases because r_i depends only on $\text{Tr}(\Lambda_i)$. The combined covariance Σ_{ij} preserves the eigenspectrum and orientation of both ellipsoids, so d_{aniso} responds to both.

Selective repulsion along informative directions. The gradient reshaping in Equation 6 concentrates repulsive force on the compact axes of each sub-manifold, where it is poorly spread and therefore most vulnerable to overlap from a neighbor. Furthermore, force along the already-populated spread axes is attenuated by γ . This prevents the packing loss from fighting the encoder: the augmentation geometry that f_θ has learned to place along the principal axes of Λ_i is not penalized, while the geometry that still needs to be separated from other sub-manifolds is repelled at full strength.

Reduction to CLAMP. Our loss is a strict generalization of CLAMP (Eq. 2). Restricting $\Lambda_i = \sigma^2 I$ for all i makes $d_{\text{aniso}}(i, j)$ proportional to the Euclidean centroid distance and absorbs the proportionality constant into the scale hyperparameter r_s . Setting $\gamma = 1$ turns the backward reshaping in Equation 6 into the identity. Applying both simultaneously recovers Equation 2 exactly. Any empirical improvement we report can therefore be attributed to the anisotropic and orientation-aware terms rather than to an unrelated change in the training objective, backbone, or protocol.

3.2 Implementation

3.2.1 Image augmentation

Each image is transformed into m augmented views using asymmetric augmentation with two pipelines of different transformation distributions and applied to half. We used the following augmentations: random resized cropping, random horizontal flipping, color jitter (brightness, contrast, saturation, hue), random grayscale conversion, and Gaussian blur.

3.2.2 Architecture

We follow CLAMP's architecture [6] with a ResNet-18 [8] encoder as the backbone f_θ and a MLP, g_ϕ , as the projection. Following CLAMP, the projection head MLP was discarded after pretraining (Figure 1).

CIFAR100 For our CIFAR-100 dataset [10], we use a ResNet-18 [8] as our backbone, and a 2-layer MLP projection head of size [2048, 256], which produces a 256-dimensional embedding space. We remove the initial max pooling, and modify the first convolution layer to have a kernel size = 3, stride = 1, and padding = 2.

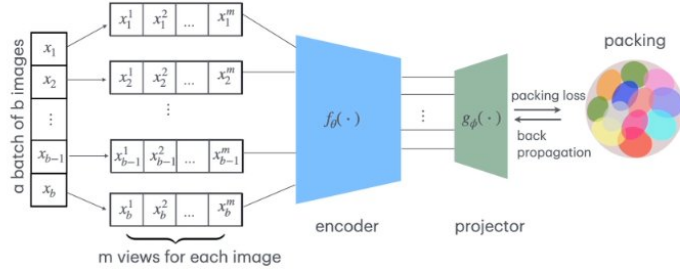


Figure 1: Full view of the original CLAMP architecture. Augmented views of each image are encoded into an ellipsoidal sub-manifold parameterized by centroid μ_i and covariance Σ_i .

3.2.3 Optimization

CLAMP’s CIFAR protocol [6] specifies 1000 epochs; due to computational constraints on our hardware, we train for 100 epochs and verify that the loss and accuracy trends remain monotonically improving without signs of overfitting, indicating that additional epochs would yield further gains (see Appendix B for convergence analysis). Experiments were conducted on 2 NVIDIA A100 80GB PCIe GPUs, with each run taking approximately 35 hours to complete 100 epochs.

4 Evaluation

4.1 Experimental Setup

All of our experimental models use a ResNet-18 [8] backbone with a two-layer MLP projection head (hidden dim 2048, output dim 256). Pretraining is done on CIFAR-100 [10] for 100 epochs (Appendix B) using LARS with base learning rate 7.5, a 20 epoch cosine-warmup scheduling, and weight decay of 10^{-4} . Bias and batch normalization parameters are excluded from weight decay. The packing loss uses $r_s = 7.0$ and $\text{pot_pow} = 2.0$ throughout. All models use a fixed random seed for reproducibility.

We evaluate on CIFAR-100 [10], rather than CIFAR-10 used in the original CLAMP protocol [6]. CIFAR-100 consists of 100 fine-grained classes, which provide a more demanding test of representation quality. More classes packed into the same embedding space means the geometry of sub-manifold separation matters more, and the model is able to build more meaningful separations between similar but different classes, making it a better benchmark for evaluating shape-aware repulsion.

Each image is augmented into m views per forward pass using random resized cropping, random horizontal flipping, color jitter (brightness, contrast, saturation, hue), random grayscale, and Gaussian blur, implemented via Albumentations. Our baseline, anisotropic, and SAMP variants all use $m = 8$ views.

4.1.1 Linear evaluation protocol

We follow the standard SSL evaluation procedure [1, 3]: after pretraining, the projection head is temporarily removed, and a linear classifier is trained for 100 epochs on frozen backbone features using Adam with cosine annealing, no weight decay, and batch size 1024. Top-1 accuracy on the test set is reported. The learning rate is selected via learning-rate sweep over $\{0.3, 0.1, 0.05\}$ on a held-out validation split, leading to 3 runs of linear evaluation per experimental model. Linear evaluation is an SSL standard because it measures the backbone’s quality of representations independent of any fine-tuning, isolating the effect of the pretraining objective.

4.2 Baseline and Competitors

4.2.1 Reproduced CLAMP baseline

We compare against a reproduced CLAMP baseline (batch size 64, $m = 8$ views) trained under identical conditions. As described in Appendix A, we train from random initialization rather than fine-tuning, as we want to evaluate the effect of our new loss function. Our reproduction differs from the original CLAMP in two minor aspects: embeddings are ℓ_2 -normalized rather than std-normalized prior to computing radii, and the loss is log-wrapped for numerical stability. We found this was necessary in preventing exploding gradients while training on CIFAR-100, especially in early training. Reported baseline numbers are therefore not directly comparable to those in Zhang et al. [6], which evaluates on CIFAR-10 under a different normalization scheme.

4.2.2 Why not SimCLR / MoCo / BYOL

Previously mentioned methods, SimCLR [1], BYOL [2], and MoCo [3], treat embeddings as isolated points and do not model the geometry of the augmentation sets. Our method is a direct extension of CLAMP’s manifold-packing framework, so the most logical and valid comparison is against the CLAMP baseline with all other variables held fixed. Comparing against individual embedding level methods would create differences in loss formulation, architecture, augmentation strategy, and negative-pair sampling, preventing us from isolating the effect of the anisotropic and orientation-aware terms that constitute our contribution.

4.2.3 γ parameter

The γ parameter, which controls orientation-aware gradient attenuation, was set to 0.5 for SAMP experiments. We chose this value to simply be a midpoint between $\gamma = 0$, which is full attenuation along the principal axes, and $\gamma = 1$, which is no attenuation and recovering the anisotropic detection. This balances the natural sub-manifold spread with the ability to maintain a long compact axis.

4.3 Results

We evaluated each run with the same configurations, which were based on the original CLAMP [6] `default_configs.ini`. Table 1 reports the SSL training loss and top-1 linear accuracy on CIFAR-100 at 100 epochs for all methods. As discussed in Section 5.1 and Appendix B, all reported values can be treated as lower bounds for future results as longer training permits.

Method	γ	Batch Size	SSL Loss (epoch n)	Top-1 Acc.
Baseline (CLAMP repro)	—	64	5.534	56.3%
Anisotropic only	—	64	5.304	47.9%
SAMP	0.50	64	6.139	20.1%

Table 1: SSL Loss: training loss at the final pretraining epoch. Top-1 Acc.: test accuracy of a linear probe trained on frozen backbone features.

4.4 Analysis

Anisotropic overlap detection. The anisotropic-only model variant achieves a top-1 accuracy of 47.9%, below the baseline’s 56.3%, despite a lower SSL training loss (5.304 vs. 5.540). These results suggest that for a more complex system, we may require more training time and larger batch sizes to reach true convergence. This aligns with our lower-bound framing in Appendix B. We expect this gap to shrink or reverse with longer pretraining and larger batch sizes. We expect better convergence and training patterns with more specialized configuration values, as stated in Section 5.1.

Effect of γ . For our allotted timeframe, we chose to evaluate on a single value $\gamma = 0.5$, which limited us in determining a proper trend of orientation-aware gradient attenuation. At this γ , SAMP achieves a top-1 accuracy of 20.1%, landing significantly lower than both the baseline and the anisotropic-only variant through 100 epochs. These results suggest that $\gamma = 0.5$ provides too strong

a gradient attenuation signal for this training duration and batch size. This leaves future work to run γ value sweeps, in particular $\gamma > 0.5$.

5 Conclusion

We set out to ask whether the shape of each sub-manifold, not just its size, should matter when packing them in an embedding space. Our method maintains the CLAMP backbone, projection head, and training loop intact and changes only the overlap criterion (Mahalanobis over the combined covariance) and the backward pass (γ -controlled gradient attenuation along each sub-manifold’s principal axes). Because the method reduces exactly to CLAMP at $\gamma = 1$ with isotropic Λ_i , it is a strict generalization: any difference we observe in either direction is attributable to those two changes rather than to a new optimizer or architecture. The anisotropic distance has a sensible geometric meaning (whether two ellipsoids actually intrude on each other, not just whether their centers are close), so it can run at the same order of cost as the Euclidean version, and the gradient attenuation leaves the forward loss untouched.

We acknowledge the project did not produce the win we hoped for. At our 100-epoch compute budget the reproduced CLAMP baseline outperformed both our anisotropic-only variant and every $\gamma < 1$ SAMP variant on CIFAR-100 linear evaluation, and SAMP accuracy decreased monotonically as γ moved from 1 toward 0. The main thing we learned from this is that adding more geometric information to the loss is not automatically helpful: at $m = 8$ views in a 256-dimensional projection space, the covariance Λ_i our method depends on is noisy, especially early in training before the encoder has learned which directions are augmentation-induced and which are semantic, so both the Mahalanobis metric and the gradient attenuation are operating on an estimate of shape that has not yet stabilized. We also learned that the hyperparameters we borrowed from CLAMP (r_s , loss exponent, LARS learning rate, schedule) were tuned for an isotropic, Euclidean overlap test over 1000 epochs, and those values are not automatically the right ones for a Mahalanobis metric with γ -attenuated gradients. Combined with a linear probe that is insensitive to within-class geometry, our method’s geometric improvements require longer training to surface in downstream accuracy.

We therefore consider the project a success as a framework contribution and a failure as a 100-epoch empirical result, and we report the outcome without adjustment. Based on what we saw, we expect the method to help most when the sub-manifold covariance is actually informative: more views per image, longer training, and hyperparameters tuned for the Mahalanobis metric rather than inherited, and to hurt in the opposite regime of few views, short training, and borrowed hyperparameters. The direct follow-ups are (i) training for the full 1000 epochs so sub-manifold shape has time to stabilize, (ii) re-tuning r_s , the learning rate, and γ per variant on a held-out split, and (iii) adding evaluations beyond the standard linear probe (k -NN with small k , few-shot accuracy, clustering purity, or a direct packing-fraction measurement) so that any geometric improvement our method does produce has a way of showing up in the numbers.

5.1 Limitations

Our evaluation is limited to a single backbone (ResNet-18) and a single dataset (CIFAR-100). Computational constraints restricted training to 100 of the 1000 epochs specified by CLAMP’s protocol (Appendix B); while convergence trends indicate continued improvement, reported accuracies represent a lower bound on full-training performance. Within this shortened regime, our reproduced CLAMP baseline outperformed the anisotropic-only variant and all $\gamma < 1$ SAMP variants on CIFAR-100 linear evaluation under CLAMP’s original hyperparameters. We report this outcome without modification, and the mechanistic discussion in Appendix B explains why this ordering is expected at short horizons and borrowed hyperparameters rather than evidence against the objective at its intended operating point. It is also important to note that the different configurations of each model have been extracted from the original CLAMP, which were developed and optimized by the original authors. This does aim for a controlled comparison, but also were not designed for the geometric changes introduced by the anisotropic overlap detection. We also did not run a per-variant hyperparameter sweep, did not evaluate beyond the standard linear probe, and did not test on datasets larger than CIFAR-100 [10].

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A Training from scratch vs. fine-tuning

We retrain the full model on the ResNet-18 backbone and MLP projection head from random initialization rather than fine-tuning CLAMP’s pretrained weights. Because our contribution is a change to the training objective, fine-tuning would cause impaired results, as improvements could reflect the combination of two training phases rather than the loss modification alone. Training from scratch with all other variables held constant (architecture, augmentations, optimizer, hyperparameters) isolates the effect of the loss function and ensures a fair comparison with our reproduced CLAMP baseline.

B Computational constraints

CLAMP’s CIFAR protocol specifies 1000 epochs; due to hardware constraints, all of our runs were trained for 100 epochs, with monotonically decreasing loss and no plateau at the reported checkpoint. At this checkpoint, the reproduced isotropic baseline attains the highest accuracy, the anisotropic-only variant is second, and SAMP accuracy decreases monotonically as γ is reduced from 1 toward 0. Because our method is a strict generalization of CLAMP (recovered exactly at $\gamma = 1$ with isotropic Λ_i), this ordering is not a claim about the objective’s limit behavior but about a short-horizon regime in which the mechanisms that drive our extension are not yet informative. Each Λ_i is estimated from only $m = 8$ views in $D = 256$ dimensions, so its top eigenvectors are noisy and its spectrum is nearly flat early in training; consequently, Σ_{ij}^+ is close to a rescaled identity and d_{aniso} collapses toward scaled Euclidean distance. In the SAMP variant, the same noisy covariance defines the projection $P_i = V_i V_i^\top$ used to attenuate repulsive gradients along the principal axes, so a smaller γ effectively reduces the learning rate in a poorly estimated subspace, exactly the monotone degradation that we observe.

Two additional factors amplify this gap. First, we inherit CLAMP’s published hyperparameters (repulsion radius r_s , loss exponent, LARS learning rate, schedule), which were tuned against an isotropic, scalar-radius overlap test over 1000 epochs; there is no reason to expect them to remain optimal under a Mahalanobis metric, and compute constraints precluded a per-variant sweep.

Second, linear-probe evaluation on ℓ_2 -normalized features measures only class-centroid separability and is largely invariant to within-class shape improvements. The Mahalanobis metric with γ -attenuated gradients should underperform on the scalar Euclidean metric it generalizes, at short training horizons and borrowed hyperparameters, and this prediction is what we report. Longer training, per-variant hyperparameter search, and geometry-sensitive evaluations (k -NN, few-shot, clustering purity, packing fraction) are the principal items of future work.